

Define frailty and discuss the role of prehabilitation for patients with frailty. (FANZCA 2019A)

Frailty is a clinical phenotype characterised by elderly patient with decreased physiological reserves greater than expected from advanced age alone.

It was described by *Fried et al* in 2001 as the presence of ≥ 3 of the following:

- Unintentional weight loss ($>4\text{kg/year}$) or sarcopenia (low muscle mass)
- Weakness (poor grip strength)
- Impaired cardiopulmonary function
 - Self reported exhaustion
 - Slow walking speed
 - Low physical activity

Other assessment tools include *Edmonton Frail Scale (EFS)* and *Frailty Index*.

Prehabilitation is a multi-modal, multi-disciplinary team approach involving the geriatrician, surgeon, anaesthetist, nurse educator, dietician, physiotherapist, psychologist, and patient to enhance patient's functional capacity to withstand a forthcoming stressor i.e major surgery.

It involves comprehensive geriatric assessment (CGA) with individualised intervention plan.

- Medical optimisation of patient's pre-existing conditions, including identifying and correcting pre-op anaemia, smoking and alcohol cessation
- Nutritional support to correct sarcopenia by increasing protein ($>1.5\text{g/kg/day}$) and total caloric intake.
- Physical exercise consisting of (1) strength training, (2) aerobic exercise and (3) flexibility exercise. Aims to improve strength, cardiopulmonary fitness and flexibility.
- Psychological support to reduce stress/anxiety. Screening for cognitive impairment, and provision of cognitive prehabilitation.

Prehabilitation has the potential to benefit patients with frailty by:

- Allowing previously non-surgical candidates to be fit enough to be considered for surgery
- Integration into ERAS protocols to
 - a. Reduce pre-op anxiety and improving patient satisfaction
 - b. Improve post-op pain
 - c. Shorten hospital length of stay
 - d. Facilitate earlier recovery of post-op function and mobility

Limitations of prehabilitation include:

- Time requirement of programme, which typically range between 4-8 weeks. This may not be feasible for patients going for emergency or time-sensitive (e.g. cancer) surgeries.
- Resource intensive nature, requiring close coordination from multi-disciplinary teams and patient commitment. Patient with poor social support may find it difficult to comply with the protocol and multiple appointments.
- Despite emerging evidence, there is no consensus on the ideal prehabilitation strategy (type and duration of intervention).
- Current literature is inconsistent and heterogenous in terms of (i) surgery type and (ii) intervention implemented, which makes it challenging for widespread adoption of a standardised protocol which is cost-effective to both patients and hospitals.